E-Valuating Seattle Charging

Aiden Tan, Katelyn Vu, Roxanne Dimadi, Jason Cao

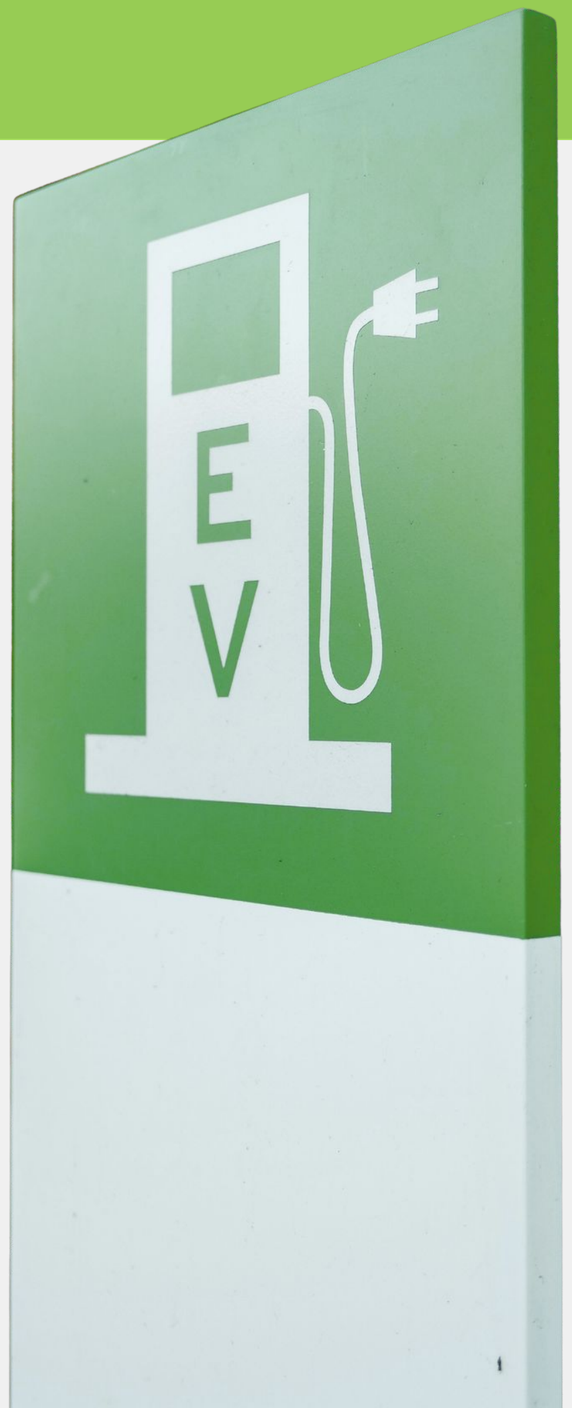
Table of Contents:

1 . Introduction and Data

2. Use Cases & Demos

- Mapping Features
- Evaluation Metric Ideation
- Predicting Placement ML Model

3. Reflection



01

Introduction

Introduction: Where should EVs belong in Seattle?

Background: Electric Vehicles are a key component to reducing transportation emissions and meeting climate goals. To support the continuous adaptation of these vehicles, charging infrastructure needs to strategically grow.

Target Audience:

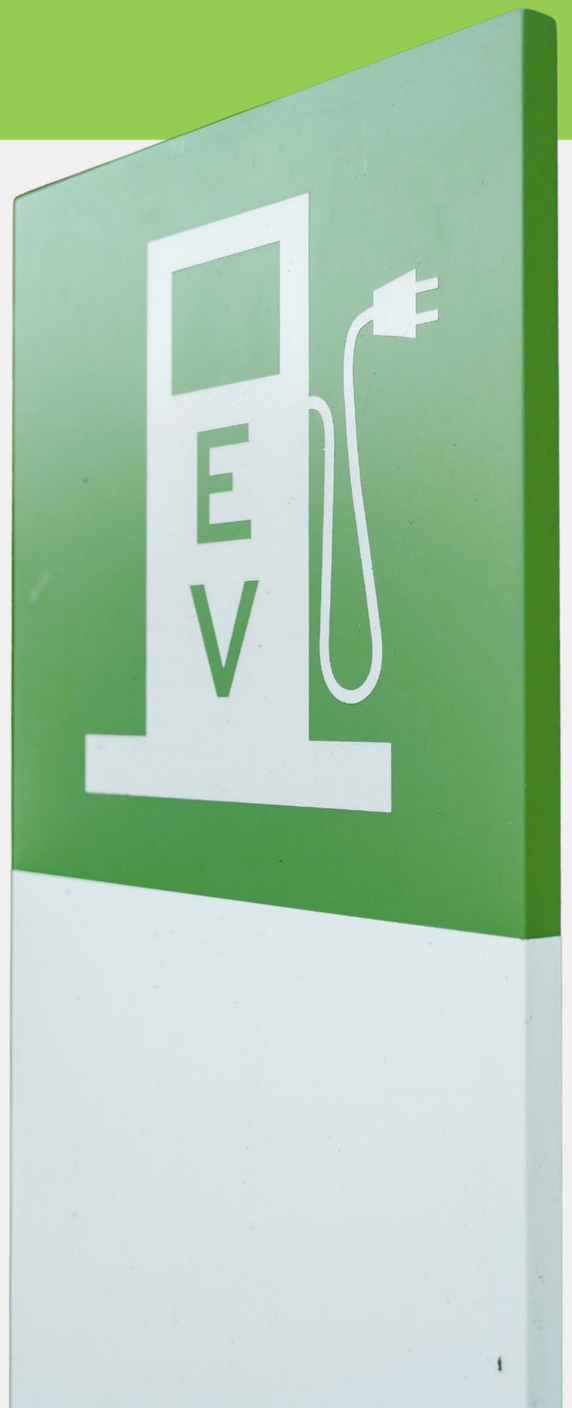
- City planners
- EV Charging

Our application:

- Maps current chargers in Seattle
- Enables custom criteria filtering
- Maps predicted chargers

Previous Work:

[Charging into inequality: A national study of social, economic, and environment correlates of electric vehicle charging stations]



Data

City of Seattle

Traffic Flow Data

STATE OF WASHINGTON OPEN DATA

EV Registration Data

Seattle City Light

Electrical Line Location Data

AMERICAN COMMUNITY SURVEY
U.S. CENSUS BUREAU

Population Data

U.S. DEPARTMENT of ENERGY

EV Charger Data

02

Use Cases & Demo

Use Cases

Use Cases

Exploring EV charging station placement across Seattle

Evaluate quality of placement of current stations

Predicting optimal locations of charging stations

Core Question

Where are the EV charging stations in Seattle, and how does EV infrastructure address existing traffic?

How effective are the placements of EV chargers based on traffic, population, and EV registration factors?

Based on the gathered data, which areas in Seattle would be optimal for EV chargers to be placed?

Web Design Components

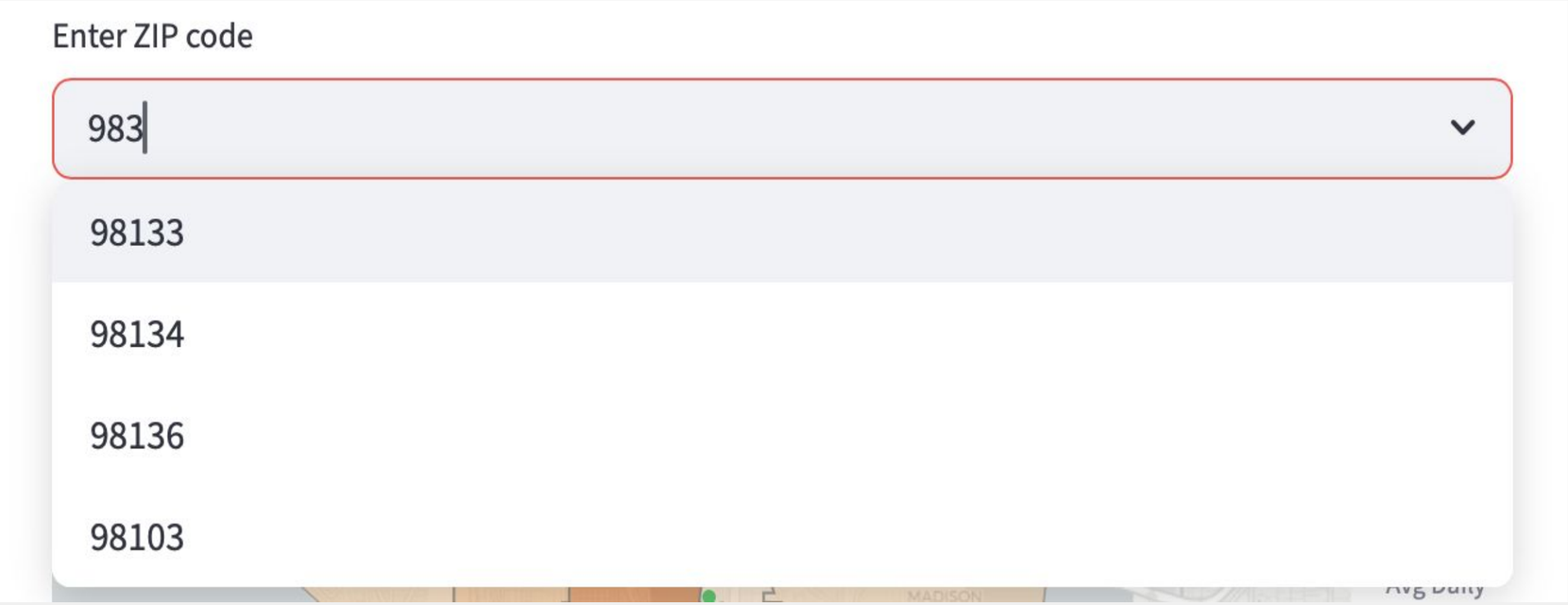
- **ZIP Code Search (Input)**
- **Road-Level Detail Map**
- **Statistics Table for selected ZIP**

- **Scoring Weights (Input)**
- **Overview Map**
- **Statistics table**

- **Maps**
- **Input data**

Use Case - Exploring EV charging station placement across Seattle

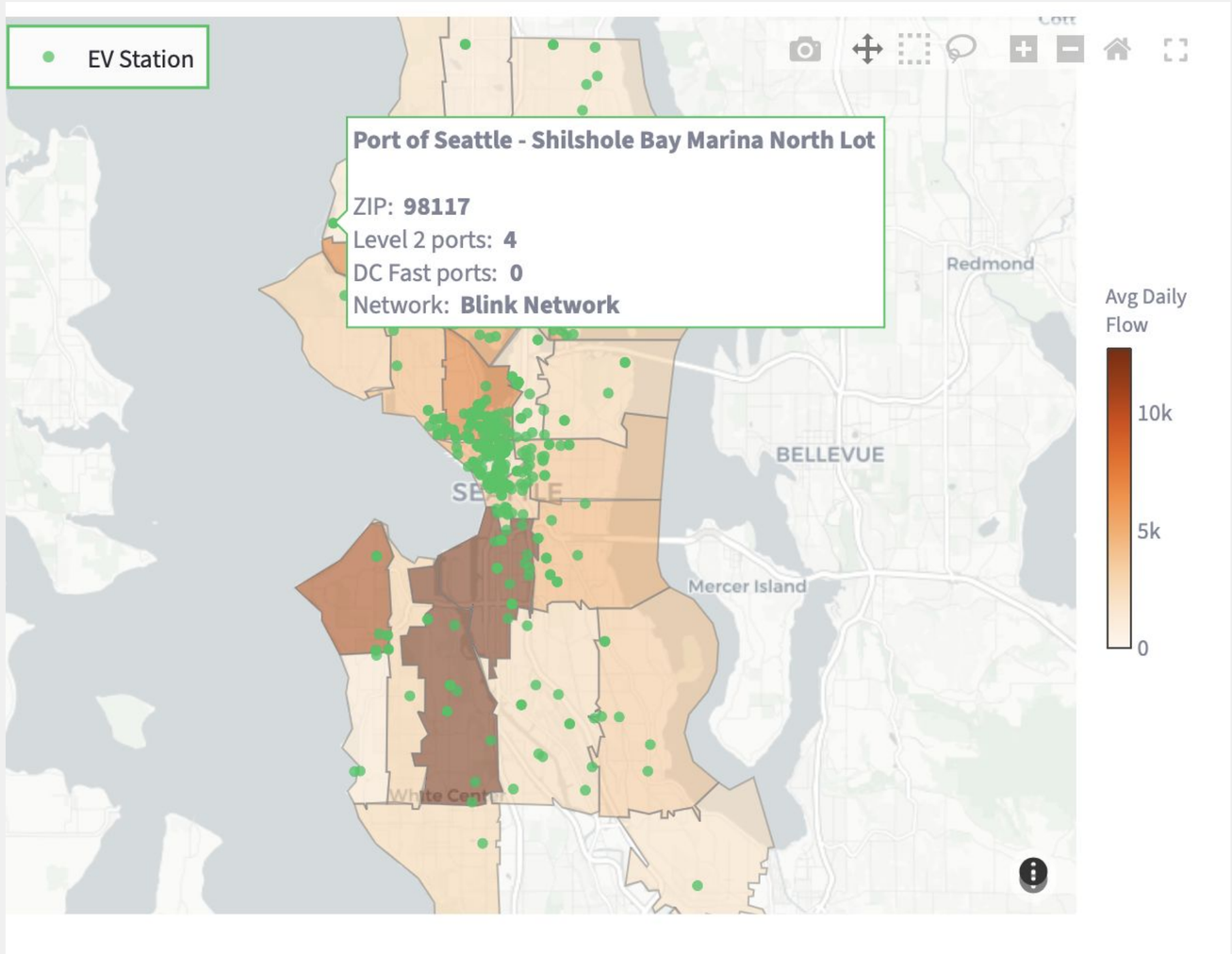
1. ZIP Code Search



- Searchable dropdown with autofill
- 29 Seattle ZIPs

Use Case - Exploring EV charging station placement across Seattle

2. Overview Map



- Each zip- code area is colored by average daily traffic flow
- Green dots showing all existing EV charging station locations
- Supports click-to-select directly on the map

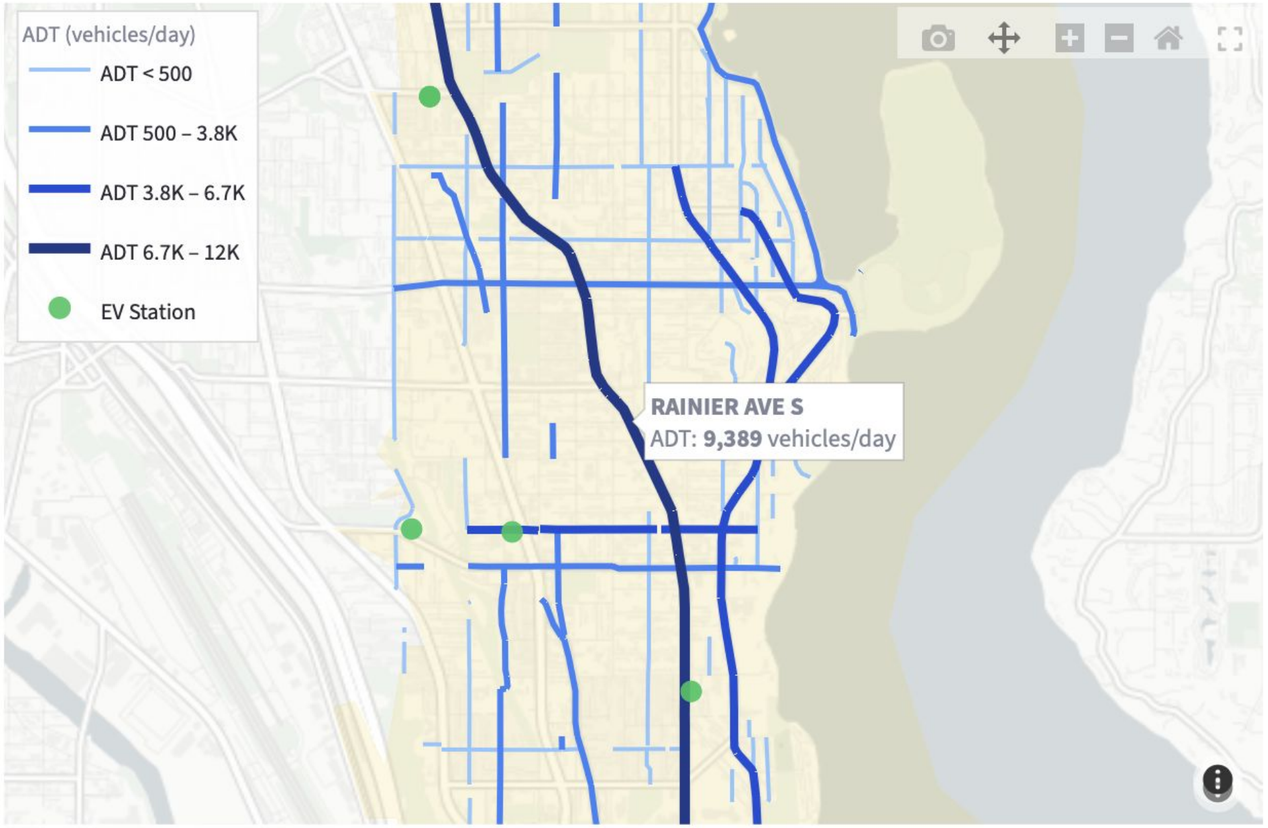
Use Case - Exploring EV charging station placement across Seattle

3. Road-Level Detail Map

SEATTLE · ZIP 98118

× Clear

Road traffic volume (ADT – Average Daily Traffic)



4. Statistics Table for selected ZIP

Population	Stations	Level2 Spots
50,500	8	14

DC Fast Spots	Avg Daily Flow
6	3,774

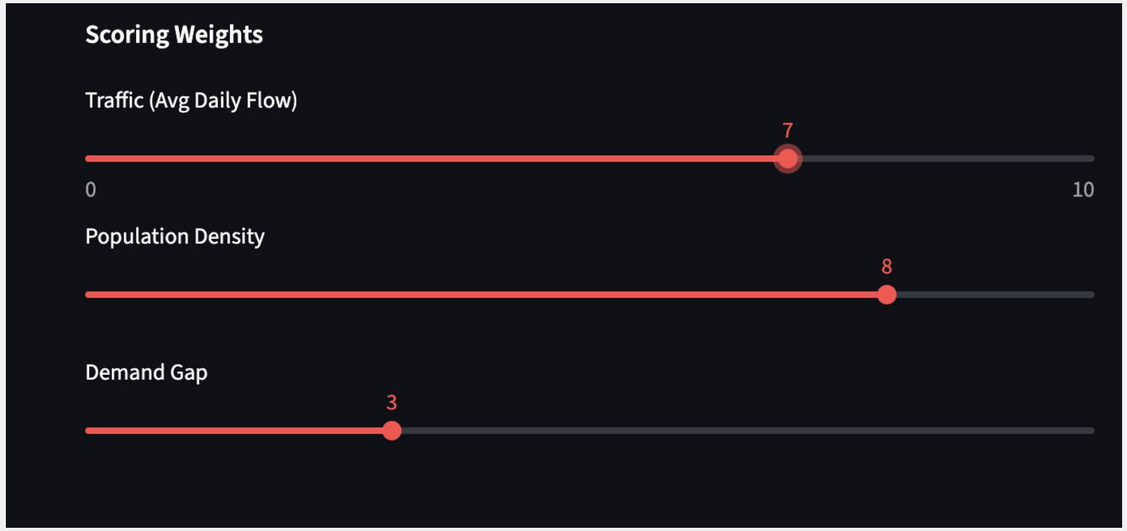
Stations in ZIP 98118

	Station	Level2	DC Fast	Network
0	Rose II Apartments	6	0	EVGATEWAY
1	Willow Crossing	2	0	Blink Network
2	SHA - New Holly - Public	2	0	OpConnect
3	CITY OF SEATAC CITY HALL 5	2	0	ChargePoint Netwo
4	CITY LIGHT COLUMBIA CITY 1	1	0	ChargePoint Netwo

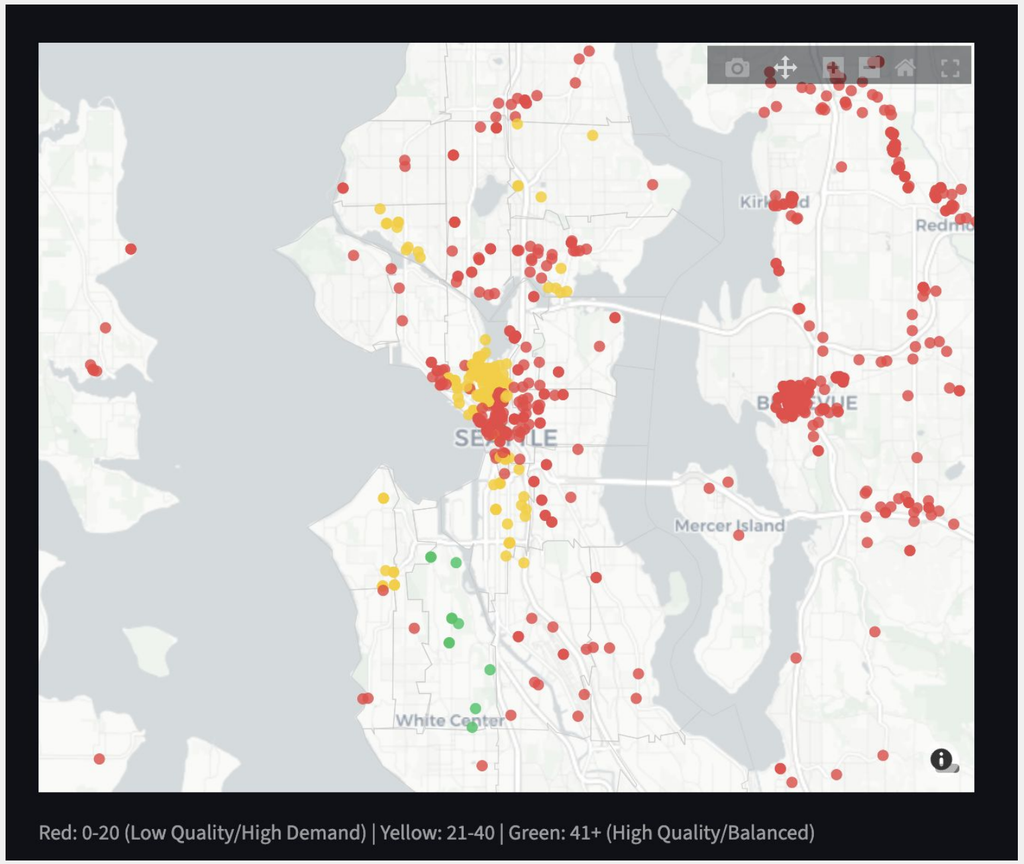
Demo - Exploring EV charging station placement across Seattle

Use Case - Evaluating quality of placement for current stations

1. Scoring Weights (Input)



2. Overview Map



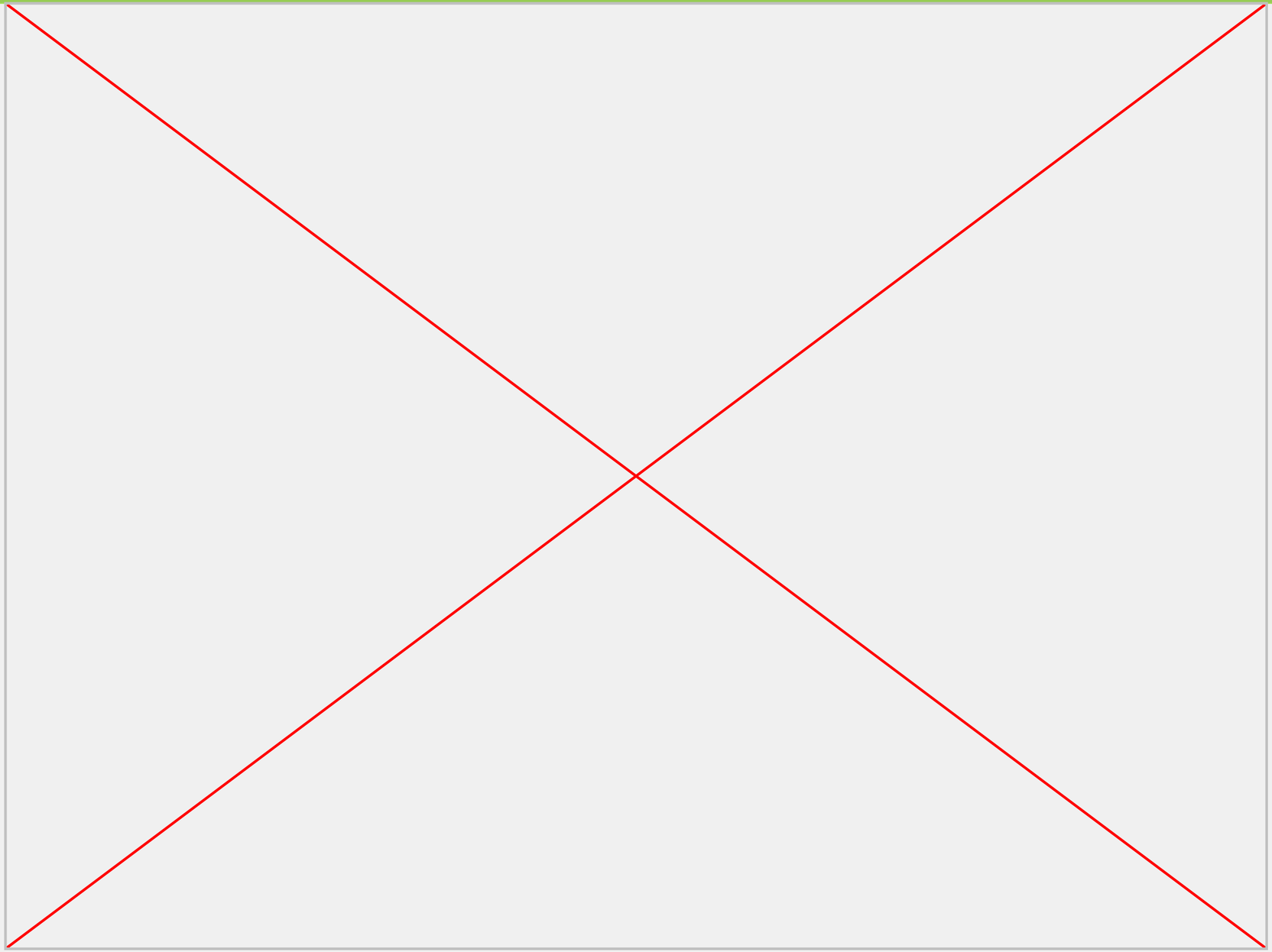
Use Case - Evaluating quality of placement for current stations

3. Statistics table

Station Scores by ZIP

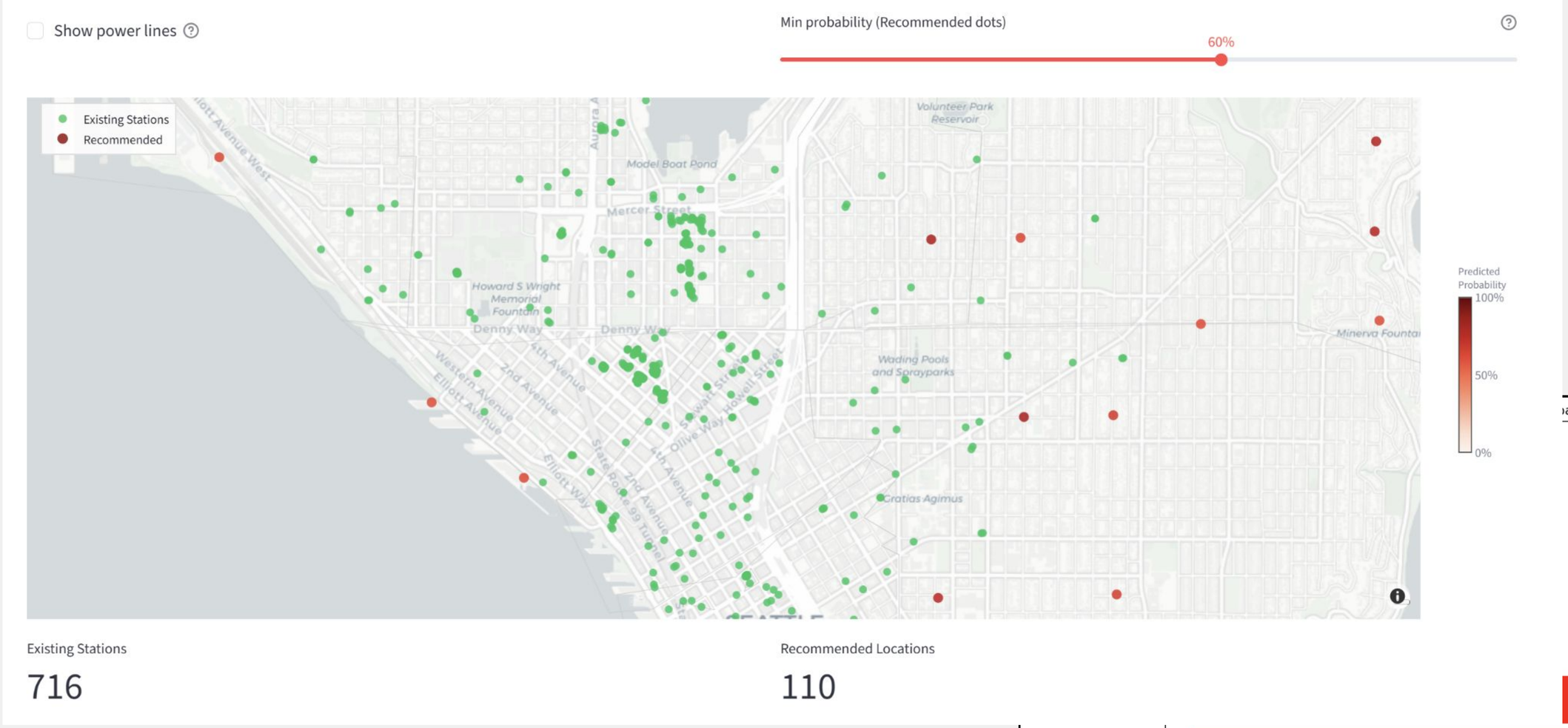
	ZIP	City	Avg Daily Flow	Population	Density (sq/mi)	Stations	Score
0	98164	SEATTLE	5136.35	292	165127.3806	13	55.3
1	98106	SEATTLE	14004.5283	28690	4940.492	11	40.8
2	98134	SEATTLE	14014.0556	645	173.3944	26	39
3	98116	SEATTLE	10798.6812	28711	9446.8523	9	32.9
4	98121	SEATTLE	5584	21166	56466.4401	178	27.5
5	98115	SEATTLE	7656.2911	54349	6448.5426	5	26.1
6	98195	SEATTLE	6729.4894	2757	30517.3025	5	24.1
7	98109	SEATTLE	7901	34328	14212.2644	156	23.4
8	98107	SEATTLE	7571.1475	28776	12018.4261	10	22.1
9	98177	SEATTLE	2408.5	21090	3704.2622	1	19.9
10	98126	SEATTLE	3480.3889	22975	7550.5549	1	19.7
11	98103	SEATTLE	6188.4824	52671	9991.7886	24	17.3
12	98146	SEATTLE	3019.75	28573	6175.4253	1	15.3
13	98178	SEATTLE	2173.2143	26748	4576.0832	1	12.8
14	98144	SEATTLE	4992.0896	33787	6508.7087	11	12.8

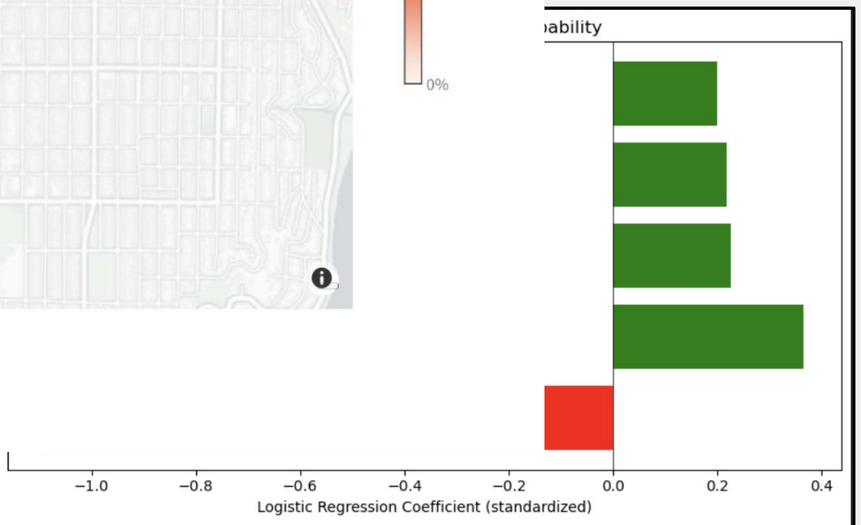
Demo - Evaluating quality of placement for current stations



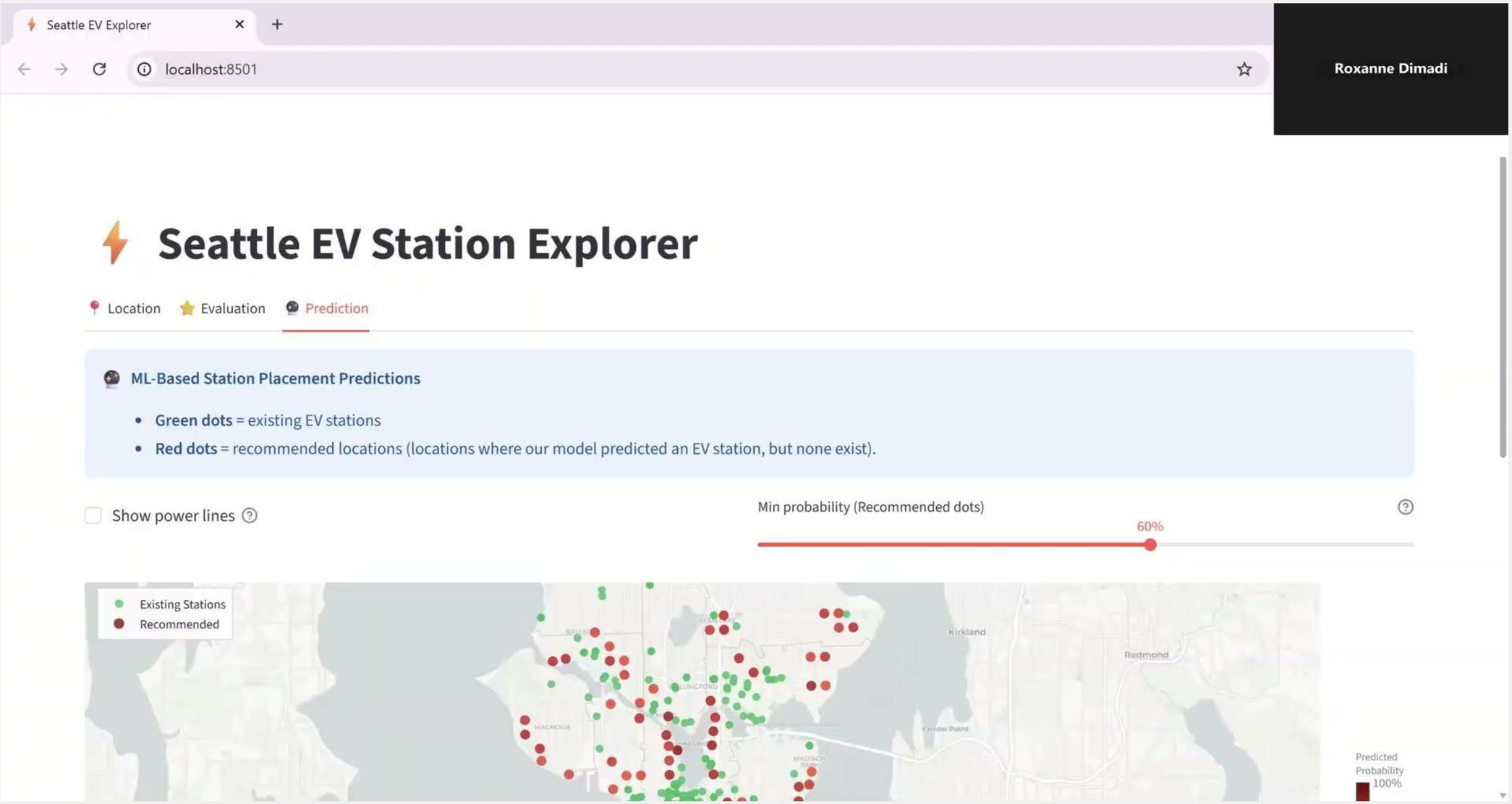
Use Case - EV Placement Recommendation

Logistic Regression





Demo - EV Placement Recommendation



03

REFLECTION

Lessons Learned

Software

- Using GitHub:
  - Dealing with pull requests
- Writing Continuous integration and tests
- Writing TOML files, YAML files.

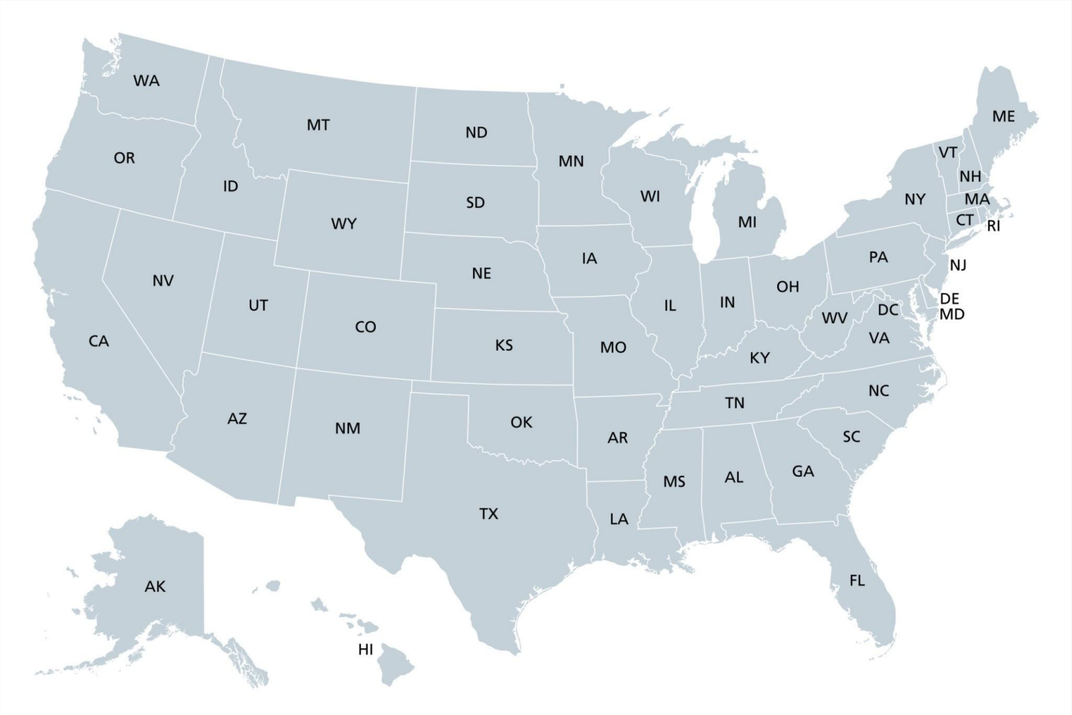
Design Elements

- Component interactions are important to identify early in the project
  - Clarifies dependencies
  - Team collaboration

Future Work

1. Evaluation Scope

2. Geographic Scope



Thank you!